

THE MILLENNIUM PROBLEMS

BY

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RECONNAISSANCE

THE CORPUS, THE STATE OF THE FIELD, AND WHAT THIS TOOLKIT REACHES

OR IOR

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*For Megan, my loving wife, the only person that really understands
what I mean when I speak.*

*For Autumn and Opal, without you this thought would have never
occurred.*

To everyone else, I wish you well.

ABSTRACT

This is reconnaissance and states no result. What the Millennium Prize problems ask, in the words of their own official statements. What became of one of them in September 2026. Which of the rest this project's apparatus could be aimed at. And the work nobody here has done. No problem gets attacked in these pages, and nobody claims priority over anything.

Seven documents make up the corpus. Five are Clay Mathematics Institute problem descriptions, covering P versus NP, the Hodge conjecture, the Riemann hypothesis, quantum Yang–Mills theory, and the Birch and Swinnerton-Dyer conjecture. The other two are Riemann: his 1859 memoir in the Wilkins translation, and four photographed pages of his handwriting. Somebody opened each one. The photographs turn out to hold a draft, not the published memoir, and no text can be pulled from them at all, though the filename says otherwise. Seed a citation off that filename and the registry inherits the error. The corpus chapter exists for that reason.

Navier–Stokes is missing from the corpus because the Clay index of unsolved problems no longer carries it. OpenAI published a finite-time blowup proof for the forced three-dimensional incompressible equations on 8 September 2026, with a Lean 4 formalization beside it. Fefferman asks for a proof of any one of four statements. Two of those four put a smooth external force inside the statement. Forcing sits in the problem as posed, then. Twice this work insisted otherwise before anyone opened the document, and the last chapter carries both retractions along with whatever killed them.

Six points went into checking that formalization against the problem statement. Viscosity strictly positive, with nothing taken to a vanishing limit. Decay conditions on the data and on the force, neither relaxed. A force smooth across the whole closed half-line. A conclusion quantified as a negated existential. Kinetic energy held bounded for all time, the start and every moment after it. And the periodic pressure condition that arrives in the errata and not in the body. All six match.

Whose statement is it, though. Not the authors' own. They took it from a Lean formalization the Formal Conjectures project had already published, kept it unmodified, and left its deliberate placeholder gaps in place, while a separate module discharges the identical statement and prints out what axioms it leaned on. A proof assistant checks a deduction and never a translation. That is this project's standing objection to machine-checked proof, and the architecture above is built to answer it.

The gaps matter as much. They sit in the sentences that carry the claims and not in a footnote. Nobody built the Lean development. Nobody machine-checked anything here. The source text was read. Of the paper, the statements and the bridging argument, nothing beyond. The proof tree stayed shut. Nobody looked into the priority dispute running through this field, and no line here should be read as taking a side in it.

Which problem can the apparatus reach? Birch and Swinnerton-Dyer, on a recommendation this research paper records without adopting. The reasoning: the refined conjecture is an exact numerical identity. Verifying it comes down to whether a computed real number lands on an integer. A precision engine answers questions shaped like that one. Riemann goes in the same

chapter as the trap. There the apparatus matches the subject and misses the obstruction, zeros on the critical line have been computed far past anything reachable here, and counting more of them bears on no proof. One null offered for the Birch and Swinnerton-Dyer measurement got withdrawn: it derived the bar instead of drawing it, and the drawn replacement stands in its place.

Nothing here claims a new method. Nothing here attributes one either. The literature behind these problems has gone unread in this work, and until somebody reads it that stays missing work, not a hole in anyone else's.

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WHAT THIS IS

Purpose: State what this research paper records, what authorized it, and the boundary of what it claims. **Scope:** theory/millennium/, and the corpus at Downloads/millennium/ as it stood on 2026-09-11.

1.1 THE RESEARCH PAPER

Written on 2026-09-11, from a corpus already gathered. The aim was narrow: check one specific claim, then write down what was found. This record came of that.

Four things sit in it. First the corpus, and what each document turns out to be once you open it, as against what its filename promises. Then the state of the Navier–Stokes problem, which moved three days before anyone started writing. Then a reading of that move against the official problem statement and against the Lean formalization that shipped with it. Last, a recommendation about where the project’s apparatus might be aimed. It is written down here and not adopted.

No problem gets attacked. Nobody measured anything, ran anything, or opened a line of attack. Lead a research paper with a method and you have already claimed the method reaches something; no claim of that kind appears here or is supported here.

1.2 WHAT IS NOT CLAIMED

Nobody claims priority in these pages and nobody wants it. theory/crystallography says as much about its own construction. Saying it twice, here, is deliberate.

The reading is the part nobody did. Take any of the five open problems in the corpus: somebody has to go through the field behind it and set down what that field already knows. Nobody here did. Until somebody does, these pages assert no new method and credit no old one.

This research paper has a second reason to be blunt about that. A priority dispute is live in the field the Navier–Stokes chapter covers. Nobody here looked into it. No position gets taken, and no line in that chapter should be mistaken for evidence on either side. The chapter reports what two documents say. Who got there first is a question it never asks.

1.3 THE RAIL THIS WAS WRITTEN UNDER

Five rules governed the writing, and each one limits what the later chapters may claim.

Cite nothing unread. A document earns a citation here by being opened. When a fact came by report instead, the sentence carrying that fact admits it.

A secondhand fact is no second witness. Whenever a fact arrived secondhand, it was confirmed against the file or the document instead of being written down on trust. Two of those confirmations came back different, and the workbook chapter holds both.

Gaps go in the sentence making the claim. Not a footnote, not an appendix. The Navier–Stokes chapter runs a whole section on what went unverified, and marks the individual soft spots where they fall.

Withdrawn claims stay on the page, with whatever killed them. The workbook chapter holds only those. Eight claims came out over the course of this work.

Draw the bar. Never derive it. Reason a threshold out of a distribution and you inherit every variance source you forgot, all of them pushing the same way. That rule turns up here as a correction and not as a precaution: the null offered for the recommended measurement had been derived, it was pulled, and the toolkit chapter carries the drawn replacement.

1.4 WHOSE DECISIONS THESE WERE

Scope never belonged to this research paper. Which problem, whether the effort exists at all, whether any of this got written: those were Douglas Quigg’s to settle. The toolkit chapter’s recommendation is set down for a ruling and not acted on.

One structural call went deliberately untaken. The shared preamble already declares theorem, lemma, proposition and definition environments, numbered within chapter, and not one existing research paper uses them; they put propositions in prose and number them by hand. Switching to the declared environments would reach all seven research papers through that one shared file. This research paper follows the practice it found and leaves the convention unchanged.

THE CORPUS

Purpose: Record what each document in the corpus is, as established by opening it, and separate that from what its filename says. **Scope:** Downloads/millennium/ as it stood on 2026-09-11, seven files, 7,852,267 bytes.

2.1 WHAT IS IN IT

Seven documents. Each was opened and identified from its own first page and embedded metadata, not from its filename. Five are the official Clay Mathematics Institute problem descriptions. Two are Riemann.

File	Identified as	Pages	Bytes
pvsnp.pdf	Stephen Cook, <i>The P versus NP Problem</i>	12	176,204
hodge.pdf	Pierre Deligne, <i>The Hodge Conjecture</i>	5	146,894
riemann.pdf	E. Bombieri, <i>Problems of the Millennium: the Riemann Hypothesis</i>	11	159,267
yangmills.pdf	Jaffe and Witten, <i>Quantum Yang–Mills Theory</i>	14	195,976
birchswin.pdf	Andrew Wiles, <i>The Birch and Swinnerton-Dyer Conjecture</i>	5	142,908
Wilkins-translation.pdf	Riemann 1859, <i>On the Number of Prime Numbers less than a Given Quantity</i> , translated by David R. Wilkins	10	119,504
riemann1859.pdf	Not the 1859 paper. See below.	4	6,911,514

2.2 THE CORPUS IS COMPLETE AND WAS NOT SHORT A FILE

Six Millennium problems remain open and the corpus holds five official statements. The natural first reading is that Navier–Stokes is missing. That reading is wrong and this work made it before checking.

The Clay Mathematics Institute index of Millennium Prize problems, read on 2026-09-11, lists five problems as unsolved – Birch and Swinnerton-Dyer, Hodge, P versus NP, Riemann, and Yang–Mills and the mass gap – and one as solved, Poincaré. Navier–Stokes appears in neither list. The

corpus holds exactly the five the index lists as unsolved. It matches the index and is not short anything.

That statement is a fact about how a website categorized its own contents on one date at one URL, and it is recorded here as that and not as a fact about the status of a problem. A site can be mid-edit, deliberately cautious, or behind. The Navier–Stokes problem’s own page, read the same day, carries the status label *Active*. The two readings are not consistent with each other, both are recorded, and neither is treated here as evidence about the mathematics.

2.3 THE DOCUMENT THAT IS NOT WHAT IT IS NAMED

`riemann1859.pdf` is not Riemann’s 1859 memoir. It is a photographic scan of Riemann’s handwritten working manuscript.

The evidence is in the file. Four pages carry zero extractable characters between them, measured page by page. The embedded document title is the scanner operator’s own filesystem path, `Cod_Ms_Riemann_3.jpg` under a directory named for the Riemann *Nachlass* and a subdirectory named *Rechenseiten*, calculation pages. The producer chain is a JPEG passed through PDF24 Creator and a Quartz PDF context in 2015, which is a scan-to-PDF path and not a typesetting one. All four pages were rendered and read as images. They are German cursive in ink, heavily struck through with interlinear corrections, foxed and water-stained, with *Nachlass Riemann* and the year 1859 legible in the header of the first page. The zeta function, the factor $\Pi(s-1)$, the logarithmic expansion of the Euler product, the function $\xi(t)$ and the logarithmic integral $\text{Li}(x)$ are identifiable in the hand.

So the file is a primary source in the most literal available sense, and it is unusable for text search or for quotation without somebody who reads nineteenth-century German *Kurrentschrift*. The content of the published memoir is in the Wilkins translation instead.

This matters to the citation registry and not only to a reader. The registry seeds an entry with empty bibliographic fields and holds them empty until somebody looks the source up. An entry seeded from this filename would name Riemann’s published 1859 memoir and point at a manuscript draft, and the empty-field convention would not catch it, because the error is in the identification and not in the completeness. The distinction has to be written into the entry.

2.4 AN OBSERVATION ABOUT PRODUCTION

Five of the six official statements share a producer string of MiKTeX pdfTeX-1.21a and creation timestamps on 2026-08-04 falling within three minutes of one another. Bombieri’s Riemann statement is older tooling, a dvips and Acrobat Distiller chain dated 2000-08-30, and carries the embedded title `Riemann-new.dvi`.

The Navier–Stokes statement, fetched separately from the Clay website for the next chapter, carries the same producer and a timestamp of 16:46:57, nineteen seconds after the Birch and Swinnerton-Dyer file. It is part of the same batch.

This is a real observation about how the statements were produced and it says nothing whatever about their contents. It is recorded because it establishes that the corpus and the separately fetched statement are the same edition of the same set, which is worth knowing in a registry entry. It is worth nothing in an argument about mathematics and is not used as one.

NAVIER–STOKES, AND WHAT A FORMALIZATION CHECKS

Purpose: Record what the official problem asks, what was published against it on 2026-09-08, and the result of checking the published formalization’s statement against the problem statement point by point. **Scope:** Fefferman’s Clay problem description, 6 pages; *OpenAI 2026 Navier-Stokes*, 166 pages; *OpenAI 2026 Euler*, 57 pages; the Lean 4 repository `openai/NavierStokesAndEuler`, four source files of 2,669 read, one of them the adapted *Formal Conjectures* statement.

3.1 WHAT THE PROBLEM ASKS

Fefferman’s official description sets up the incompressible equations on $\mathbb{R}^n$ for $n = 2$ or 3 , with viscosity $\nu > 0$, and imposes decay and regularity conditions numbered (4) through (11). It then states, in the sentence that governs everything after it:

To give reasonable leeway to solvers while retaining the heart of the problem, we ask for a proof of one of the following four statements.

The four are (A) existence and smoothness on $\mathbb{R}^3$, (B) the same on $\mathbb{R}^3/\mathbb{Z}^3$, (C) breakdown on $\mathbb{R}^3$, and (D) breakdown on $\mathbb{R}^3/\mathbb{Z}^3$. In (A) and (B) the force f is taken identically zero. In (C) and (D) it is not. Statement (C) reads, in his words, that for $\nu > 0$ and $n = 3$ there exist a smooth divergence-free u° on $\mathbb{R}^3$ and a smooth $f(x, t)$ on $\mathbb{R}^3 \times [0, \infty)$, satisfying (4) and (5), for which there exist no solutions (p, u) of (1), (2), (3), (6), (7) on $\mathbb{R}^3 \times [0, \infty)$. Statement (D) is the same with (8) and (9) on the data and (10) and (11) on the solution.

A smooth external force is therefore named inside two of the four statements, and the problem asks for a proof of any one of the four. The only occurrence of the word *prize* in the document is a remark that the Euler equations are not on the Clay Institute’s list. There is no passage reserving the problem for (A) or (B).

The conditions that matter for the check below are (4), rapid decay of every derivative of the initial data; (5), rapid decay of every space-time derivative of the force on the closed half-line; (6), smoothness of p and u on $\mathbb{R}^n \times [0, \infty)$; and (7), $\int |u(x, t)|^2 dx$ bounded by one constant for all $t \geq 0$, which Fefferman annotates *bounded energy*.

The document carries errata and one of them adds a condition. The first reads that the further condition $p(x + e_j, t) = p(x, t)$ should be made explicit in Eqn (8). The periodic problem therefore requires a periodic pressure, which the body of the statement does not say.

3.2 WHAT WAS PUBLISHED AGAINST IT

On 2026-09-08 OpenAI published *Finite time blowup for Navier–Stokes*, 166 pages, together with a companion Euler paper and a Lean 4 repository. Its Theorem 1.1 is a construction, not directly a non-existence statement:

For every $\nu > 0$ there exist a force $f \in C_c^\infty(\mathbb{R}^3 \times (0, 1); \mathbb{R}^3)$, a compact set $K \subset \mathbb{R}^3$, and smooth velocity and pressure fields u, p on $\mathbb{R}^3 \times [0, 1)$ satisfying $\partial_t u + (u \cdot \nabla)u - \nu \Delta u + \nabla p = f$, $\nabla \cdot u = 0$, $u(\cdot, 0) = 0$, such that $\text{supp } u(\cdot, t) \cup \text{supp } p(\cdot, t) \subseteq K$ for every $0 \leq t < 1$, $\sup_t \|u(t)\|_{L^2(\mathbb{R}^3)} < \infty$, and $\limsup_{t \rightarrow 1} \|u(t)\|_{L^\infty(\mathbb{R}^3)} = \infty$.

Constructing one solution that blows up does not by itself establish (C). Statement (C) asks that *no* smooth bounded-energy solution exist on the whole half-line for that datum and force, and a blowing-up solution leaves open whether some other smooth global one exists. The paper closes that gap and says so in the sentence after the theorem, asserting that there is then no smooth solution on $\mathbb{R}^3 \times [0, \infty)$ with the same force and initial datum whose kinetic energy is uniformly bounded, and that this establishes alternative (C) as stated by Fefferman.

The argument for the gap is in the proof of Theorem 1.1 in section 10.4. Smoothness and fixed compact support put the field in $C([0, T]; H^3)$ for each $T < 1$; the embedding $H^3(\mathbb{R}^3) \hookrightarrow L^\infty(\mathbb{R}^3)$ excludes a classical H^3 continuation through time one; a difference-energy calculation with Grönwall gives uniqueness on every shorter interval, making $[0, 1)$ the maximal classical existence interval. A supposed global smooth competitor of uniformly bounded energy is then forced by the comparison lemma to agree with the constructed field throughout $[0, 1)$, which contradicts its own boundedness. Every fixed positive viscosity is reached by a spatial rescaling that leaves the singular time at one.

Statement (D) is Corollary 10.6, obtained by rescaling the compact supports into the interior of a fundamental cube and summing integer translates, whose supports stay disjoint so the nonlinear term survives. The corollary’s proof states that the pressure is periodic as well, *as required in the erratum to the problem statement*. The authors read the errata.

3.3 WHAT A PROOF ASSISTANT DOES NOT CHECK

This project’s position on machine-checked proof, stated before this chapter was written, is that Lean verifies the deduction and never the claim. A proof assistant confirms that a conclusion follows from the assumptions as written. It cannot confirm that the statement formalized is the statement anybody meant. A machine-checked proof of a neighboring theorem compiles exactly as cleanly as a machine-checked proof of the right one, and nothing in the compile distinguishes them. The burden therefore moves entirely onto the formalized statement, and that statement is a translation. No proof assistant checks a translation.

That makes the formalized statement a finite document a person reads word by word. The rest of this chapter is that reading.

3.4 THE STATEMENT CHECK

Six points were checked against the Lean statement in `ComparatorChallenges/NavierStokes.lean` and against Fefferman’s numbered conditions. The first five were specified before the file was opened. The sixth was added because the file itself raised it.

Point	What the Lean statement carries	Verdict
Viscosity strictly positive, and not a vanishing-viscosity limit	$(\nu : \mathbb{R}) (\text{h}\nu : \nu > 0)$, universally quantified over the theorem	matches
Condition (4) on the initial data, unrelaxed	<code>InitialVelocityConditionDecay</code> : divergence-free, C^∞ , and for every derivative order m and every K a constant C with $\ \nabla^m u_0(x)\ \leq C/(1 + \ x\)^K$	matches
Condition (5) on the force, unrelaxed, on the closed half-line	<code>ForceConditionDecay</code> : C^∞ on $\text{univ} \times [0, \infty)$ and $\ \nabla^m f(x, t)\ \leq C/(1 + \ x\ + t)^K$ for all $t \geq 0$	matches
Conclusion in the negated existential form Fefferman quantifies	<code>not (exists v p, NavierStokesExistenceAndSmoothnessRn nu u0 f v p)</code>	matches
Energy bounded uniformly in time, not at $t = 0$ alone	<code>globally_bounded_energy</code> : there exists E such that for all $t \geq 0$, $\int \ v(x, t)\ ^2 < E$; plus square-integrability at each $t \geq 0$	matches
The periodic pressure condition from the errata	<code>isOnePeriodic_pressure</code> on the periodic structure, annotated in the source as following the Clay errata	matches

The equation itself is encoded with the time derivative taken as a derivative within $[0, \infty)$. Fefferman states equation (1) on the closed half-line, and the Lean encoding follows him there. Divergence is defined as the trace of the Jacobian and carries an explicit junk value of zero at points of non-differentiability, which the source declares instead of leaving implicit.

3.5 THE STATEMENT IS NOT THE AUTHORS’ OWN

The file carrying the statement is a *comparator challenge*. Its header records that it was copied, at a named commit, from the Formal Conjectures project’s Lean formalization of the Navier–Stokes problem statement, that the breakdown alternatives (C) and (D) are retained *including their intentional placeholder gaps*, and that neither the proof root nor the submission imports it. Both theorems in that file end in a placeholder and prove nothing. It is a yardstick.

The discharge is a separate module. It restates both theorems with textually identical statements and closes each by applying a bridge lemma from the project’s own development, declaring no axioms and leaving no placeholder, and it ends by printing the axiom dependencies of both theorems. The repository’s instructions direct a checker to verify the two challenges with an external

checking tool under a sandbox, using an independent exporter and an independent kernel instead of Lean’s own.

That architecture is a direct answer to the objection this chapter opened with. The concern is that an author who writes both the statement and the proof can machine-check a translation of their own choosing. Here the statement was written by a third party, published beforehand, for the purpose of stating the problem and not of being proved; the authors adopted it unmodified and discharged it; and the axiom base is printed for a reader to check. The translation is still the thing to read, and this chapter read it, but the party that wrote it had no stake in what would later be proved against it.

3.6 THE UNFORCED EULER COMPANION

A second paper was published alongside, *Finite time blowup for the Euler equation*, 57 pages, and it matters here for one reason: it is unforced.

It studies the unforced incompressible Euler equations on $\mathbb{R}^3$, $\partial_t u + (u \cdot \nabla)u + \nabla p = 0$ with $\nabla \cdot u = 0$, taking initial data in the smooth compactly supported divergence-free class and writing $T(u_0)$ for the maximal lifespan of the smooth solution. Its Theorem 1.1 asserts that there exists such a u_0 with $0 < T(u_0) < \infty$, whose smooth solution satisfies $\limsup_{t \rightarrow T} \|u(t)\|_{L^\infty} = \infty$ and $\int_0^T \|\operatorname{curl} u(t)\|_{L^\infty} dt = \infty$.

The second divergence is not decoration. The Beale–Kato–Majda criterion, which the paper states and attributes, says a smooth three-dimensional Euler solution continues past T whenever that integral is finite. Finite-time breakdown therefore requires it to diverge, and the theorem delivers exactly the quantity the criterion demands. By the authors’ own account the Euler result came first, as the easier problem, and was used to prompt the Navier–Stokes effort.

This is not a Millennium result and is not recorded as one. Fefferman’s statement says the problems are also open and very important for the Euler equations at $\nu = 0$, and that the Euler equation is not on the Clay Institute’s list of prize problems. The distinction is his and is kept here.

3.7 WHAT THE UNFORCED EULER RESULT DOES NOT SETTLE

The reason to hold this paper is a question put to this work directly: what is left as the forcing goes to zero? The test behind it is whether a mechanism repairs a property of the object or a property of the setup, and a mechanism that needs the force to be there has taught us about forced systems and nothing about the unforced problem.

The Euler paper looks like the answer and is not. **Removing the force and removing the viscosity are different removals.** Statements (A) and (B), the two alternatives that remain open on anybody’s reading, are $\nu > 0$ with f identically zero. The Euler equations are $\nu = 0$. An unforced blowup at zero viscosity therefore does not transfer, and it does not transfer for the precise reason that the regularity problem exists: the viscous term is the smoothing whose sufficiency is the entire question. A construction that blows up with no viscosity to fight is not evidence about what happens when there is.

Put as the limit test, the two papers move in orthogonal directions. One removes the viscosity and keeps nothing to resist the amplification. The other keeps the viscosity and supplies a force. The open cases keep the viscosity and remove the force, and neither paper is that case.

That is an argument from the shapes of the statements and not from their mechanisms. The amplification mechanism was not read in either paper, and the honest version of the limit test requires reading it instead of reasoning from the abstract. What is recorded here is that the question is still open and that the obvious shortcut to it does not work.

3.8 WHAT WAS NOT VERIFIED, STATED HERE AND NOT IN A FOOTNOTE

Nothing here was machine-checked. The Lean development was not built. No proof was executed, no axiom output was observed, and the claim that the discharge module compiles without placeholders rests on reading its source text. The check in this chapter is a reading of four files out of 2,669.

The proof was not read. The 166-page paper was read at its abstract, its introduction, its Theorem 1.1, the section 10.4 argument that bridges to the non-existence form, and Corollary 10.6. The mathematical content of sections 3 through 9 and the three appendices was not read. The bridge lemma the discharge module applies was not traced into the development. Of the Euler paper's 57 pages, only the abstract, the introduction and Theorem 1.1 were read; its amplification mechanism, the part the limit test actually needs, was not. Nothing in this chapter rests on the correctness of either proof, and nothing in it could, because neither was read.

The priority dispute was not investigated. A dispute over credit is running in this field. It was not examined for this research paper, no position is taken on it, and nothing in this chapter bears on it either way.

One reported fact is recorded as reported. That the authors have stated they will not claim the Millennium Prize appeared in a headline encountered during the search and was not confirmed against any statement of theirs. It is written here as an unverified report and should not be repeated from this research paper as anything else.

And the status question is separate from the mathematics question. Whether the Clay Mathematics Institute has accepted, awarded, or will award anything was not verified and is not asserted. What this chapter establishes is narrower and is the part that is checkable: the statement discharged in Lean is the statement Fefferman wrote, on all six points examined.

WHAT THIS TOOLKIT REACHES

Purpose: Record the recommendation made about which problem the project’s apparatus could be pointed at, the reasoning offered for it, the trap identified alongside it, and the measured bound that constrains it. **Scope:** `PQC/examples/proofing/bbp_scaling.py`, `PQC/theory/PARTITION.tsv`.

Whose this is. The recommendation in this chapter was given on 2026-09-11 in answer to a direct question. It is recorded here and not adopted, and no work has begun on it.

4.1 THE OBVIOUS ANSWER, AND WHY IT IS THE TRAP

An engine whose edge is arbitrary precision reaches for Riemann first, and Riemann is the wrong choice for a reason that has nothing to do with how good the arithmetic is.

The apparatus matches the *subject* and not the *obstruction*. A zeta function and unbounded precision look like a fit. But computing zeros on the critical line is long done, far past anything a desktop reaches, and no finite count of zeros bears on the proof at all. Adding zeros reproduces existing work at a worse scale. The conjecture is not precision-bound. An engine whose advantage is precision buys nothing against it.

That is the superficial-match failure in its exact form, and it is worth having named because it is the shape of mistake this apparatus is most exposed to: the subject looks like the instrument. Yang–Mills wants constructive quantum field theory and lattice gauge machinery. Hodge wants algebraic geometry with no numerical handle this toolkit has. P versus NP is not a numerical question, and Douglas Quigg had already set it aside on the separate ground that the anisotropic boundary has not been worked out.

The specific figures on Riemann given for this argument were not independently checked. The claim that zeros have been computed at height 10^{22} and that verification has covered the first 10^{13} of them was supplied with the recommendation. The argument does not depend on it, because it turns on the conjecture not being precision-bound, whatever height has been reached.

4.2 THE RECOMMENDATION

Birch and Swinnerton-Dyer, on the ground that its obstruction matches the instrument, where the other Millennium problems match only in subject.

It is the only one of them found by computation: Birch and Swinnerton-Dyer fitting curves numerically on EDSAC. It is natively experimental, and the refined conjecture is an exact numerical

identity. For an elliptic curve E of rank r , writing Sha for the order of the Tate–Shafarevich group, Ω for the real period, Reg for the regulator and c_p for the Tamagawa numbers,

$$\frac{L^{(r)}(E, 1)}{r!} = \frac{\text{Sha} \cdot \Omega \cdot \text{Reg} \cdot \prod_p c_p}{|E_{\text{tors}}|^2}.$$

Rearranged, the analytic order of Sha is a quotient of computable real numbers, and the conjecture asserts that this quotient is an integer, usually a perfect square. The verification is therefore: compute a real number to N digits and ask whether it lands on an integer. That is a precision-bound question, and it is the check the engine already performs.

4.3 THE FIRST MEASUREMENT, AND THE POSITIVE CONTROL IT REQUIRES

The positive control comes first and is not optional. Curves are taken from published tables where the rank and the order of Sha are already known, at rank zero and one, small conductor, with Sha trivial or known to be 4 or 9. The quotient is computed and must land on the known integer. An instrument that cannot reproduce a known answer says nothing when pointed at an unknown one.

The quotient is only as good as its weakest input. The real period and the regulator come from their own algorithms carrying their own error. The honest report is therefore the precision of the quotient and not the precision of the arithmetic. An arithmetic with no floor under it does not give its inputs one.

4.4 THE NULL: THE DERIVED VERSION IS WITHDRAWN

A null was offered with the recommendation and then withdrawn the same day. The withdrawal is the substantive part.

Withdrawn. That a real number carrying no integrality structure has its distance to the nearest integer distributed uniformly on $[0, \frac{1}{2}]$. That landing within 10^{-N} of an integer by chance has probability about 2×10^{-N} , and agreement to fifty digits is therefore a bound of 10^{-50} .

What killed it. It was derived and not drawn. A threshold reasoned out from an assumed distribution omits the variance sources its author failed to enumerate, and omitted variance only ever pushes one way. The rule in this project is to draw the bar and never to derive it, and this null was derived in the same message that stated the positive-control rule.

One narrower objection raised against it was itself wrong. The suggestion was that the integer factors in the quotient, the torsion order squared, the Tamagawa product and $r!$, might concentrate the quotient near integers and inflate the coincidence rate. They do not: scaling a structureless real by a known rational does not concentrate it near integers. The null was wrong for the better reason and not for that one.

The drawn replacement measures the null under the same conditions as the effect, through the same pipeline and with the same inputs and the same precision losses, in two ways. Forming the quotient with the rank taken one too high or one too low gives a condition under which the identity should not hold, and the distribution of distance-to-nearest-integer across many curves is then the null itself. Computing the quotient across many curves whose Sha is known gives

the spread of the residual about the true integer. That spread is the instrument's own floor, and the period and the regulator will dominate it well before the arithmetic does. The second is the positive control and the floor measurement in one pass.

The consequence changes what may be claimed. The honest bound is not 10^{-N} for N digits of arithmetic. It is however close the pipeline gets on a curve whose answer is already known. If the regulator runs out at 10^{-12} , then 10^{-12} is the claim however many digits the multiply carries.

4.5 THE MEASURED BOUND THAT CONSTRAINS THIS, AND IT WAS CHECKED

`bbp_scaling.py` in the PQC tree measures what depth costs in the integer relation search, and its recorded results were read in the file instead of taken on report.

Cost goes as precision to the 2.64. That is the measured exponent. Predicted was 2.585, from a Karatsuba host multiply near width^{1.585} against a count linear in the width. Blame does not fall on the reduction.

Where the card overtakes the host: 1,024 limbs. Call it 32,768 bits. Mean operand runs 0.642 of the precision. The mean multiply arrives at that crossover somewhere near 51,000 binary places. One cell at 51,036 places, by the file's own table, costs 26.2 hours.

None of that binds the recommended measurement, which wants tens to hundreds of digits. Two other things it does bind. Relation detection at tens of thousands of digits is off the table. So is any plan resting on the card to rescue it. The file carries a note that the prediction ran the other way before anyone measured.

4.6 A PRECONDITION THAT HAS TO BE SETTLED FIRST

The work spans two repositories with different governance, and this was established by checking.

The arithmetic the recommendation depends on lives in the PQC tree and is classified EXTERNAL in PARTITION.tsv under a ruling quoted there from Douglas Quigg on 2026-09-11, that advantage research is held and precision work is fair. The relation search, the transform multiply and the arbitrary-precision arithmetic all carry that class. So the recommended line of attack is not blocked by the partition.

The adjacent material is blocked, and the boundary is narrow. The same file classifies the harmonic machinery as internal, under a separate ruling quoted there that anything used to derive anisotropic properties is internal. A line of attack that begins wanting that machinery has crossed into held material and stops for a ruling.

The theory research papers are in a different repository from the arithmetic, with its own custodian. Anything that wants files from both needs that settled before it starts and not after.

4.7 WHAT IS NOT CLAIMED

This is verification and not proof, and that distinction governs everything claimed in this chapter.

A measurement of this kind produces evidence for particular instances of the conjecture, in the way the original EDSAC work did. A verified instance at unusual rank or unusual precision is a real contribution and it is not a solution.

The recommendation was recorded with plain caveats: that no expertise in elliptic-curve L -function computation stands behind it, that computing $L^{(r)}(E, 1)$ requires modular symbols or Dokchitser's algorithm, that established systems already implement those and rebuilding them would be foolish, and that the realistic shape is established algorithms for the L -function with this engine supplying precision underneath. The first honest task is finding out whether that seam works at all. That assessment is recorded as theirs and is not softened here.

SOURCES, AND HOW MUCH OF EACH WAS READ

Purpose: Name every source this research paper rests on, with the registry key it resolves to and how much of it was actually read. That no claim here can be traced to a document nobody opened.

Scope: twelve sources entered for this research paper, ten of them held.

Twelve rows. Ten files. Two rows have no file behind them, and one file has no text in it.

Each row carries the key as the public tree defines it. A surname in a sentence buys a reader nothing. An author, a title, an identifier they can follow: that buys them the source. Fields absent from a document stay empty in the registry, with a note naming which. Blank misleads nobody. A guess misleads everybody.

Read-state belongs in the entry. Five of these were opened to page one and shut. Two were read in part, and the part is named. One cannot be read. Strip that out and a bibliography degrades into a list of things that exist.

5.1 READ IN FULL

Fefferman, Charles L. Fefferman, *Existence and Smoothness of the Navier–Stokes Equation*, Clay Mathematics Institute Millennium Prize Problem statement.

Six pages, errata included, cover to cover. Nothing else in the corpus got that treatment. The whole Navier–Stokes chapter turns on this one document. That earned it the reading. No year appears anywhere on it, and the paper examined in that chapter cites it without one too. The year field stays empty and the row says why. Digest for digest, its bytes match what claymath.org served on 2026-09-11.

5.2 READ IN PART, WITH THE PART NAMED

OpenAI 2026 Navier-Stokes, *Finite time blowup for Navier–Stokes*. 166 pages. Abstract, introduction, Theorem 1.1, the bridging argument in section 10.4, Corollary 10.6. Nobody counted how many pages that came to. Sections 3 through 9 and all three appendices carry the mathematics and went unread.

OpenAI 2026 Euler, *Finite time blowup for the Euler equation*. 57 pages. Abstract, introduction, Theorem 1.1. Nothing beyond that. The amplification mechanism sits in the unread remainder, and the limit test in the previous chapter wants precisely that mechanism.

Formal Conjectures, the Lean formalization of the Navier–Stokes problem statement published by the Formal Conjectures project. Read as source text, in the adapted copy the OpenAI repository carries, at a commit that file names. Four files out of 2,669. None of them built.

Everything this research paper finds about the Navier–Stokes claim rests on that one file. No copy sits under `sources/`: a repository has no single file to hold. The identifier is a path and the commit lives in the adapting file’s header.

5.3 READ AT THE FIRST PAGE ONLY

Five official Clay problem descriptions, for the five problems the index still lists as unsolved. Each was opened, identified off its own first page instead of its filename, and shut. All five match the bytes claymath.org served on 2026-09-11. Two sit under a different upload month than the rest, checked by fetching them instead of assumed from the pattern.

Cook, Stephen Cook, *The P versus NP Problem*, 12 pages.

Deligne, Pierre Deligne, *The Hodge Conjecture*, 5 pages.

Bombieri, E. Bombieri, *Problems of the Millennium: the Riemann Hypothesis*, 11 pages. Older tooling than the other four.

Jaffe and Witten, Arthur Jaffe and Edward Witten, *Quantum Yang–Mills Theory*, 14 pages.

Wiles, Andrew Wiles, *The Birch and Swinnerton-Dyer Conjecture*, 5 pages.

One page of Wiles. Wiles is also the problem the previous chapter recommends pointing the apparatus at. The shortfall bites hardest here. That recommendation stands on an exact numerical identity, and the identity did not come off this document. Whoever acts on the recommendation opens this first.

Wilkins 1998, Bernhard Riemann translated by David R. Wilkins, *On the Number of Prime Numbers less than a Given Quantity*, 10 pages. The key’s year is the translation’s, printed there as a preliminary version of December 1998. November 1859 appears on that same page for the memoir underneath, which gets its own row. English, legible, and the practical way into the 1859 content.

5.4 NO TEXT TO READ

Riemann 1859, Bernhard Riemann, handwritten working manuscript, *Nachlass Cod. Ms. Riemann 3, Rechenseiten*.

Four pages. Zero extractable characters, counted page by page. All four were rendered and read as pictures, the only way they can be read. The year on the row is the manuscript’s; the scan happened in 2015.

Its filename says published memoir. What it holds is a draft. That gap is why the entry reads the way it does, and the corpus chapter defines what seeding a row off the filename would have done to a future reader.

5.5 NOT A DOCUMENT

Clay Mathematics Institute. Two facts in this research paper describe what that website said on 2026-09-11, and a fact needs somewhere to point.

The Millennium Problems index listed five unsolved, one solved, and Navier–Stokes in neither bucket. The Navier–Stokes page itself carried the label *Active*. Those two do not agree. Both go on the record here as statements about how a website sorted its own contents on one day at one address. Neither is used anywhere in this research paper as a statement about a problem’s status.

5.6 NAMED IN THESE SOURCES AND NOT READ HERE

Reference lists carry names, and this research paper repeats a handful at second hand. Leray on global weak solutions. Caffarelli, Kohn and Nirenberg on partial regularity. The Beale–Kato–Majda continuation criterion. Buckmaster and Vicol on non-uniqueness, then Albritton, Brué and Colombo on the same. Elgindi on axisymmetric blowup. Dokchitser’s algorithm for L -function values.

Not one was read. Not one is registered. None is cited here as established.

Each gets the status of a thing a read source mentions, which is all it has earned. Writing rows for them would assert a reading nobody here has done. Somebody should do that reading. When they do, the rows are theirs to fill.

BIBLIOGRAPHY

Every measurement in this work is built on somebody's published result, and the first purpose of this chapter is that those people are credited by name. A surname in a footnote is not a citation. The second purpose is that the credit is checkable: each source is registered in `orior_citations`, where the copy sits itself, and this bibliography is generated from that registry and pinned to one commit of it. A reference therefore resolves not to a title but to exact bytes.

Credit here does not depend on agreement. Where this work reproduces a published result it says so, and where it fails to reproduce one it says that too, under the same names, because a refutation rests on the original as much as a confirmation does.

REGISTRY COMMIT 2aae1103d28d9b919eae59cc9062aeaffe1fa0b2

DATED 2026-10-01

INVENTORY MANIFEST.tsv, signed; the hashes below are its rows

A reader who disagrees with a number in this research paper can ask which copy it was checked against, and the hash answers exactly. Two copies of a paper are rarely the same bytes.

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